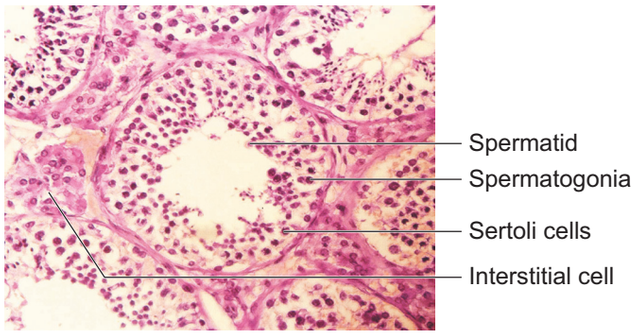


Fig. 2.6 (a) Cross sectional view of seminiferous tubule

Some of the primary spermatocytes undergo first meiotic division to form two secondary spermatocytes which are haploid with 23 chromosomes each. The secondary spermatocytes undergo second meiotic division to produce four haploid spermatids. The spermatids are transformed into mature **spermatozoa (sperms)** by the process called **spermiogenesis**. Sperms are finally released into the cavity of seminiferous tubules by a process called **spermiation**. The whole process of spermatogenesis takes about 64 days. At any given time, different regions of the seminiferous tubules contain spermatocytes in different stages of development **(Fig. 2.6 b)**. The sperm production remains nearly constant at a rate of about 200 million sperms per day.

Spermatogenesis starts at the age of puberty and is initiated due to the increase in the release of **Gonadotropin Releasing Hormone (GnRH)** by the hypothalamus. GnRH acts on the anterior pituitary gland and stimulates the secretion of two gonadotropins namely **Follicle Stimulating Hormone (FSH)** and **Lutenizing Hormone (LH)**. FSH stimulates testicular growth and enhances the production of **Androgen Binding Protein (ABP)** by the sertoli cells and helps in the process of spermiogenesis. LH acts on the Leydig cells and stimulates the synthesis of **testosterone** which in turn stimulates the process of spermatogenesis.

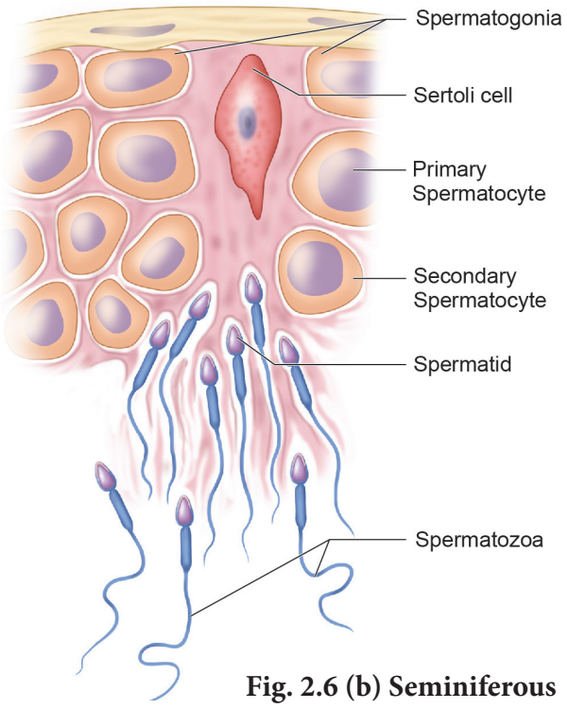


Fig. 2.6 (b) Seminiferous tubules (Enlarged)

Structure of human spermatozoan

The human sperm is a microscopic, flagellated and motile gamete **(Fig. 2.7).** The whole body of the sperm is enveloped by plasma membrane and is composed of a head, neck, middle piece and a tail. The head comprises of two parts namely acrosome and nucleus. **Acrosome** is a small cap like pointed structure present at the tip of the nucleus and is formed mainly from the Golgi body of the spermatid. It contains **hyaluronidase**, a proteolytic enzyme, popularly known as sperm lysin which helps to penetrate the ovum during Fertilization. The nucleus is flat and oval. The neck is very short and is present between the head and the middle piece. It contains the proximal centriole towards the nucleus which plays a role in the first division of the zygote and the distal centriole gives rise to the axial filament of the sperm. The middle piece possesses mitochondria spirally twisted around the axial filament called **mitochondrial spiral or nebenkern**. It produces energy in the form of ATP molecules for the movement of sperms. The

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